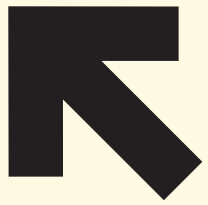


# Kokkos Comm:
## Performance-Portable Communication

C. Nicole Avans & Evan D. Suggs

# Agenda

# 3 Introduction

Kokkos Comm is a performance-portability communication API. It exists to manage error-prone implementation details and provide a core interface for switching between communication backends (e.g., MPI, NCCL, RCCL, etc.)

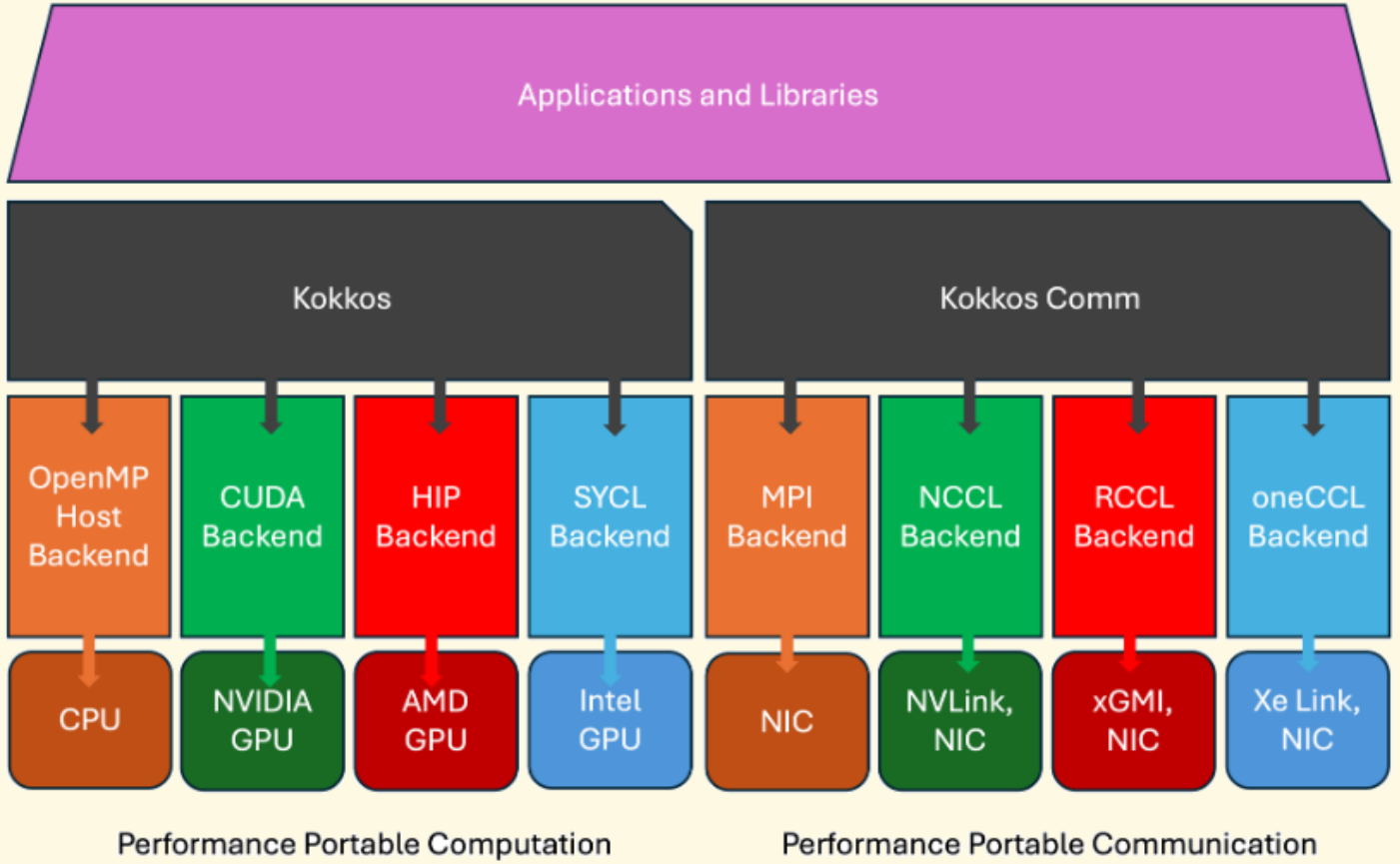
# 4 Review of Major Concepts

| | |
|---|---|
| **Kokkos** | Write-once, run-anywhere performance portability for on-node computing |
| **Kokkos Comm** | Communication abstractions for hardware-agnostic off-node communication |
| **MPI** | Message Passing Interface, used generally to facilitate off-node communication |
| **NCCL / RCCL** | Hardware-specific collective communication libraries for Nvidia and AMD GPUs, respectively |
| **OpenMP** | Parallel features for shared-memory environments |

# 5 Kokkos Ecosystem Stack

Kokkos and Kokkos Comm exist within an ecosystem of libraries, and each tackle different facets of parallelism and performance-portability.

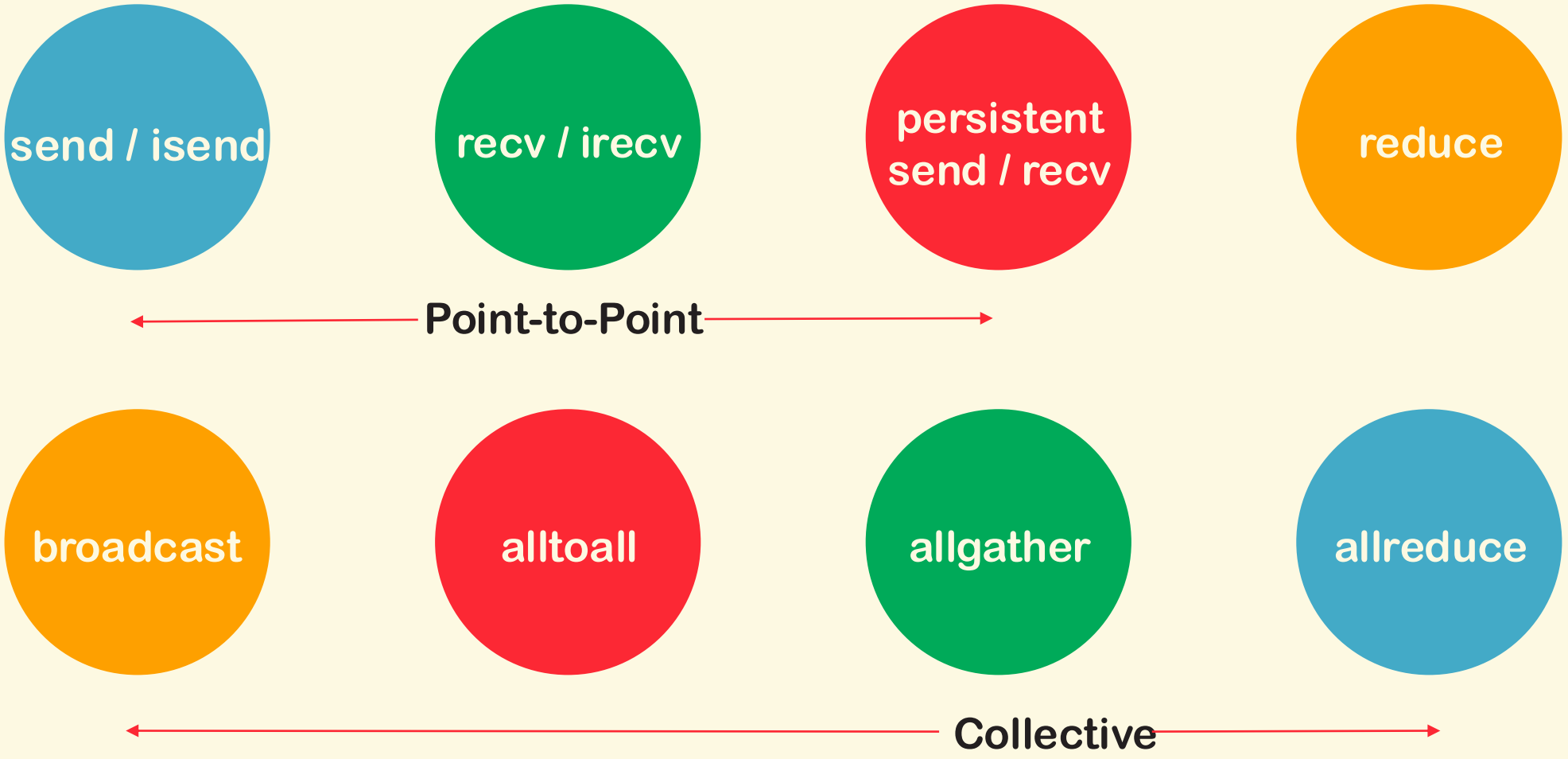
Question: How much, if any, hardware-specific C++ have you written at this point? Have you experienced any friction?



# 6 What's Error-Prone About It?

- If your data is non-contiguous, it must be marshalled (ordered), packed, sent, unpacked, then computed.
- MPI provides communication for a variety of hardware, but it deals with pointers rather than data structures.
- Communication has typically not been write-once run-anywhere: vendor-specific collective communication is written separately, and optimization requires knowledge about the architecture of the intended system.

# 7 Communication & Collectives




See the MPI Standard for exhaustive information about these and more communication patterns.

# 8 Non-blocking Send Comparison

**MPI_Isend specification**

```
int MPI_Isend(const void *buf, int count,
  MPI_Datatype datatype,
  int dest,
  int tag,
  MPI_Comm comm,
  MPI_Request *request);
```

**KokkosComm::mpi::isend specification**

```
template <KokkosExecutionSpace ES,
    KokkosView SV>
void isend(const ES &space, const SV &sv,
  int dst,
  int tag,
  MPI_Comm comm,
  MPI_Request &req);
```

**KokkosComm:send specification**

```
template <KokkosExecutionSpace
    ES, CommunicationSpace CS, KokkosView
    SV>
Request<CS> send(Handle<ES, CS> &handle,
  const SV &send_view,
  int dest);
```

**Reminder:**
**In C++, you can turn a pointer into a value with * (*buf) and turn a value into a pointer with & (&req).**

# 9 Persistent Communication

```
KokkosComm::Channel<> channel(dest_rank, src_rank, tag, MPI_COMM_WORLD);

Kokkos::View<Scalar*, Kokkos::HostSpace> send_host("send_host", N);
Kokkos::View<Scalar*, Kokkos::HostSpace> recv_host ("recv_host", N);

channel.sendinit(send_host);
channel.recvinit(recv_host);

channel.start();
channel.wait();
```

A simple persistent communication example.

# Getting Started

# 11 Aside: Build Methods

## CMake

- CMake is the de-facto standard for building C++ code.
- The –S flag specifies project root directory and –B flag specifies build directory.

sudo apt install cmake

## Spack

- Spack is a package manager for making installing scientific software easy.
- Clone it from github, run the setup script and you're ready to go in any environment.

git clone https://github.com/spack/spack.git
. spack/share/spack/setup-env.sh

Sources linked above.

# 12 Cluster Instructions Overview

- On clusters (or locally!) you may use spack to install Kokkos and proceed to installing Kokkos Comm with CMake.
  - This is similar to using module load on clusters.
- Alternatively, you can just use CMake for all of it.
  - It is a good idea to save your CMake commands to a build script (i.e., [cluster_name]_build.sh) so you can write it once and reuse it from there. Spack commands can also be put into a script. (An example will be shown later in this presentation.)

# 13 TN Tech Cluster

- Use spack load kokkos@4.7~cuda gcc@14 openmpi cmake
- You will also want to go to the directory cd /work/classes/csc4760-001-2026s/<username> and use your username.
- Additionally, use either hpcshell --account=csc4760-001-2026s --cpus-per-task=2 or salloc --account=csc4760-001-2026s --cpus-per-task=2 to go off the login node onto a node with actual resources.
- To get location of spack module (install directory), use spack location -i <spack spec> such as spack location -i kokkos@4.7~cuda to print the location
- Follow the rest of the instructions, but use cluster directories

# 14 Delta Cluster

- **Use** module load cray-mpich cmake
- **Use** accounts **command to find what account you can use, should be** bfrm-delta-gpu
- **Example:** salloc --cpus-per-task=2 --gpus-per-node=1 --account=bfrm-delta-gpu --partition=gpuA40x4-interactive,gpuA100x4-interactive

**You will have to install Kokkos and Kokkos Comm yourself — more details to follow.**

# 15 CMake Build Commands

## Kokkos

```
cmake \
    -S [source] \
    -B [build] \
    -DCMAKE_CXX_COMPILER=[compiler] \
    -DCMAKE_INSTALL_PREFIX=[install] \
    -DKokkos_ENABLE_[CUDA/HIP]=ON \
    -DKokkos_ARCH_[type]=ON \
    -DCMAKE_BUILD_TYPE=Release;
```

## Kokkos Comm

```
cmake \
    -S [source] \
    -B [build] \
    -DCMAKE_CXX_COMPILER=[compiler] \
    -DKokkos_ROOT=[kokkos install];
```

**Bracketed text indicates where specific information needs to be changed, such as providing source, build, and install directories, the location of the compiler, or where a flag needs to be modified to select an architecture and specify CUDA (Nvidia) or HIP (AMD).**

# 16 WSL / Linux

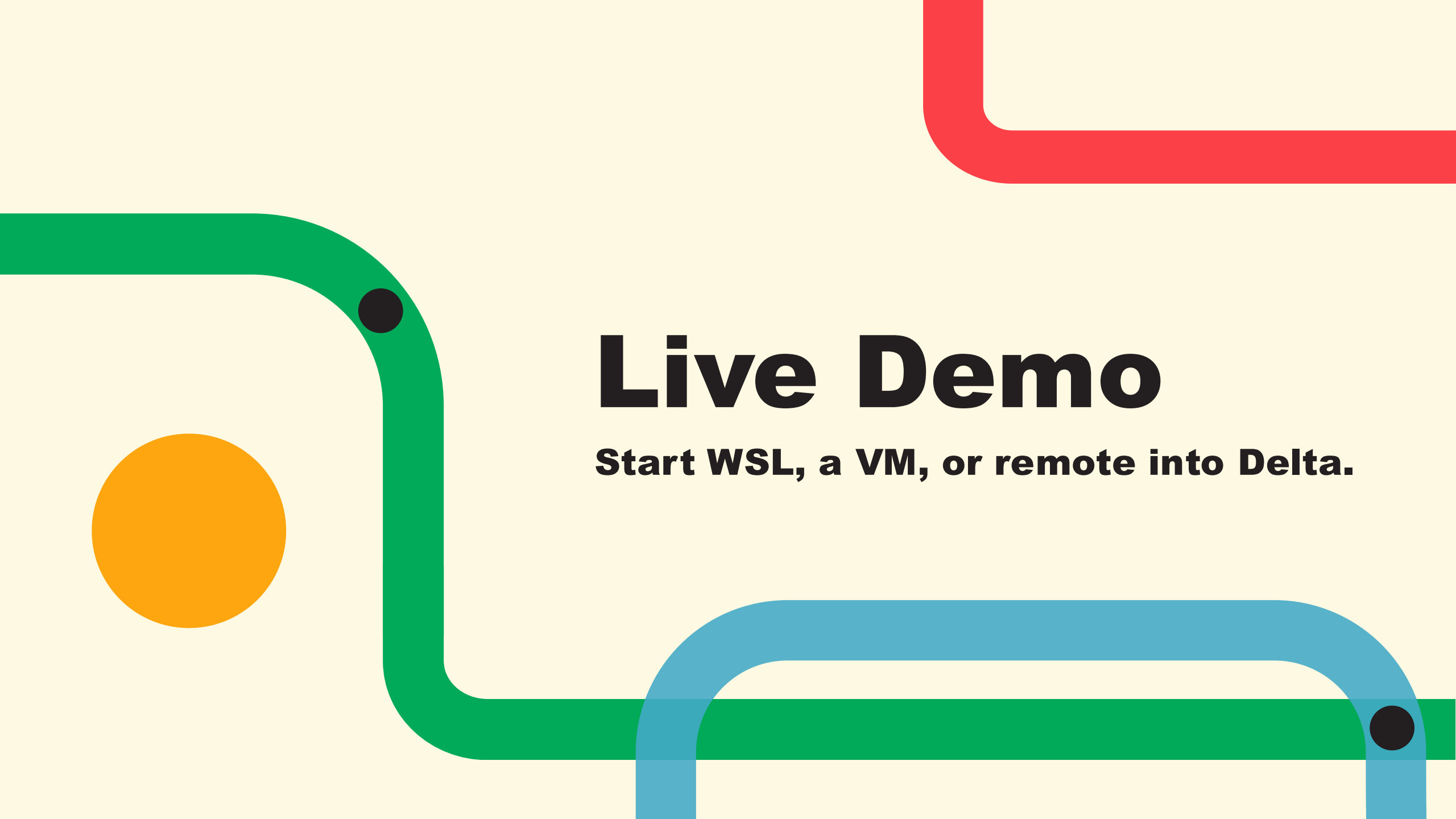
```
#!/bin/bash

init() {
   printf "Setting up directories...\n"
   mkdir -p $HOME/class/repos/ $HOME/class/install/kokkos
   printf "Complete!\n"
}

kokkos() {
   if [ -d $HOME/class/repos/kokkos ]; then
      printf "'kokkos' directory already exists.\n"
   else
      git clone git@github.com:kokkos/kokkos.git $HOME/class/repos/kokkos
   fi
   cmake \
      -S $HOME/class/repos/kokkos \
      -B $HOME/class/repos/kokkos/build \
      -DCMAKE_INSTALL_PREFIX=$HOME/class/install/kokkos \
      -DCMAKE_BUILD_TYPE=Release \
      -DCMAKE_CXX_COMPILER=mpicxx \
      -DKOKKOS_USE_CXX_EXTENSIONS=ON;
   make -C $HOME/class/repos/kokkos/build -j16 install
   printf "Kokkos setup complete!\n\n"
}
```

```
kokkos-comm(){
   if [ -d $HOME/class/repos/kokkos-comm ]; then
      printf "'kokkos-comm' directory already exists.\n"
   else
      git clone git@github.com:kokkos/kokkos-comm.git $HOME/class/repos/kokkos-comm
   fi
   mkdir -p $HOME/class/repos/kokkos-comm/build
   cmake \
      -S $HOME/class/repos/kokkos-comm \
      -B $HOME/class/repos/kokkos-comm/build \
      -DCMAKE_CXX_COMPILER=mpicxx \
      -DKokkos_ROOT=$HOME/class/install/kokkos \
      -DCMAKE_BUILD_TYPE=Release \
      -DKokkosComm_ENABLE_PERFTESTS=ON \
      -DKokkosComm_ENABLE_TESTS=ON;
   make -C $HOME/class/repos/kokkos-comm/build -j16
   printf "KokkosComm setup complete!\n\n"
}

clean() {
   printf "Cleaning up...\n"
   rm -rf $HOME/class/install/kokkos $HOME/class/repos/kokkos-comm/build
   printf "Cleanup complete!\n"
}

test(){
   cd $HOME/class/repos/kokkos-comm/build; ctest -V
}
```

```
# Check for arguments
if [ "$1" == "clean" ]; then
   clean
elif [ "$1" == "test" ]; then
   test
else
   set -e
   sudo apt install cmake
   sudo apt install openmpi-bin
libopenmpi-dev
   clean; init; kokkos; kokkos-comm; test
fi
```

**This is a complete, basic build script for WSL / Linux.**

**Question: Do you tend to use build scripts in your development? Is any of this unfamiliar?**


Live Demo
Start WSL, a VM, or remote into Delta.

# Steps

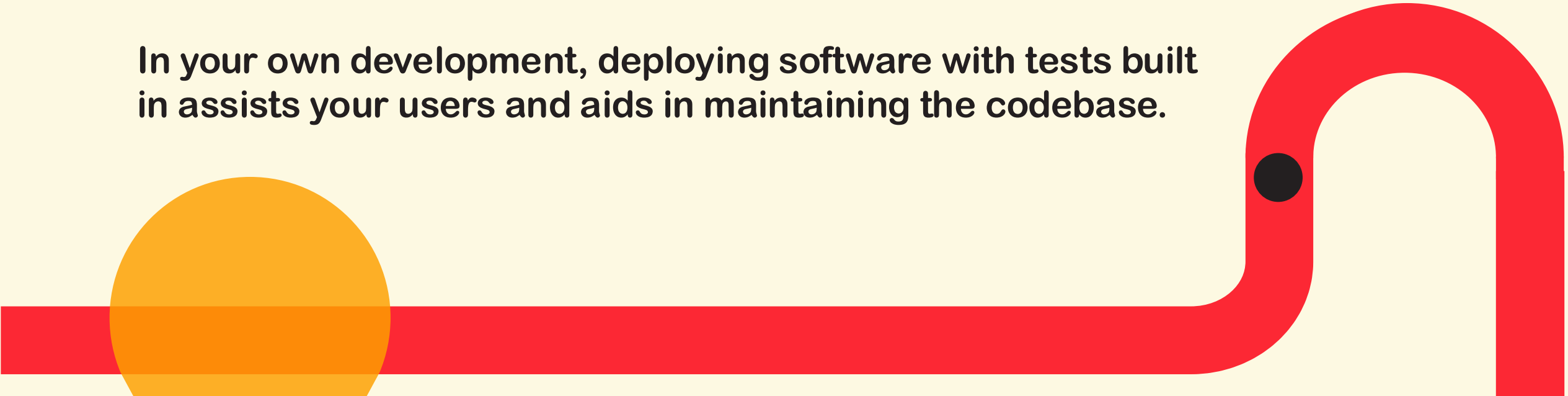
| | |
|---|---|
| **Check for MPI** | which mpi |
| **Install Kokkos** | **Or, verify current install** |
| **Install Kokkos Comm** | **Or, verify current install** |
| **Run Tests** | ctest -v |
| **Create Experiment File** | touch example.cpp |

**Don't worry – we can roll it back to previous slides to show commands again.**

# 19 Aside: Testing

CMake provides a testing utility in the form of CTest. The command shown on the previous slide (ctest –v) runs tests with verbose output. Utilizing this function with installed scientific software (such as Kokkos and Kokkos Comm) will quickly illuminate any issues running on a given system.

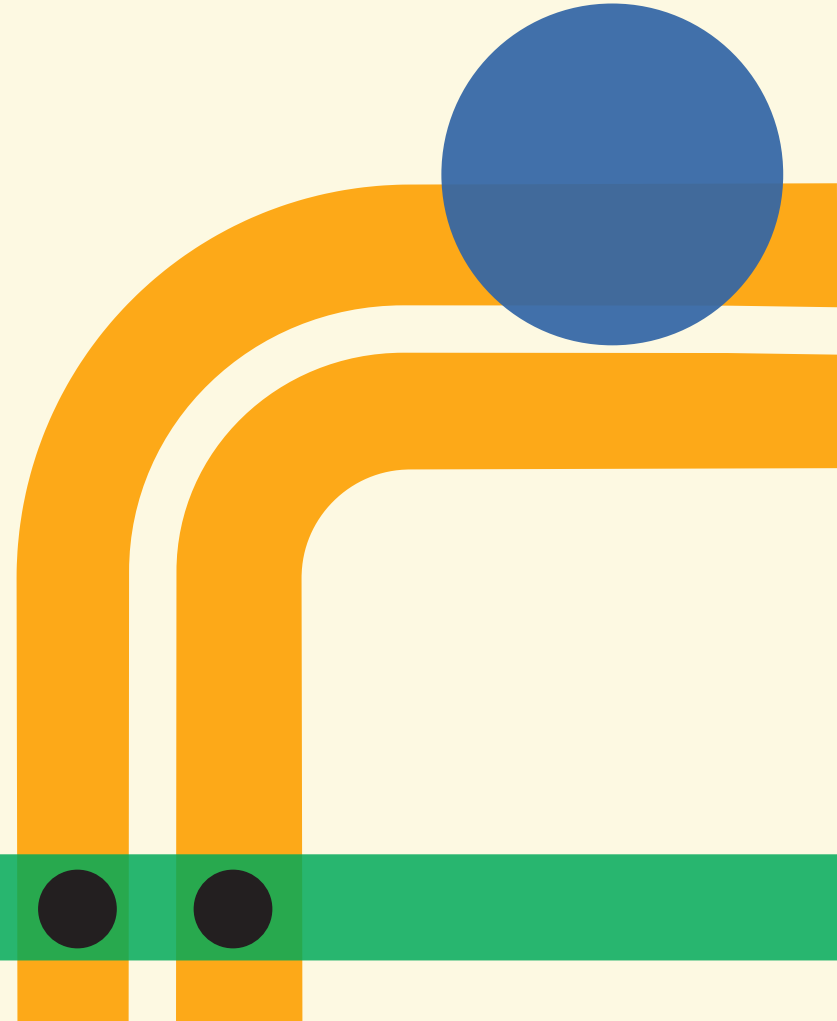
In your own development, deploying software with tests built in assists your users and aids in maintaining the codebase.



# 20 Summary

Kokkos Comm can make moving your data for parallel computation easier. In conjunction with Kokkos, it provides an enhanced experience for writing production code to be used in heterogeneous computing environments (like those found in academic, DOE laboratory, or industry clusters).

This allows you to focus on what you need to do with your data, rather than system-specific optimization or troubleshooting pointers.



# 21 Summary, cont.

This presentation intends to demonstrate basic setup and usage of Kokkos Comm within the Kokkos ecosystem, but is also an illustration of critical skills for working within scientific computing: building, testing, and continued development with integrated libraries. In particular, the use of build scripts is critical for reproducability and collaboration.

# 22 Resource Review

- Kokkos (and its list of supported architectures)
- Kokkos Comm
- MPI Standard
- CMake
- Spack (and specific instructions for use with Kokkos)
- CTest

Feel free to follow up (contact info on next slide) for more information.

# Thank you, questions?

C. Nicole Avans
cnavans42@tntech.edu
Evan D. Suggs
esuggs@tntech.edu